ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician Basic Maintainer (00379-04)

USQ Roundsperson VIC (ALVD)

OJPR MITE – 122608

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex H “Course History”.

Revision	Approval Date	Authority
1.0	18 October 2024	CNTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training Group (CNTG) and in accordance with the following references:

a. A-P9-050-000 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-004 Non-Commissioned Member General Specification, dated 09 Feb 2011;

c. A-PD-055-002/PP-002, Occupational Specifications, Marine Technician, dated 23 Mar 2018;

d. NPTORD 4500-22, Naval Training Order – Qualification Standard and Plan, dated Jun 2017; and

e. NPTORD 4500-55, Naval Training Order – On-the-Job Training Management, dated 19 Mar 2018.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFSA	All	122608	ALVD

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL TRAINING GROUP (CNTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the roundsperson on a VIC class submarine. This QSP is designed to assist training establishments in developing, conducting, evaluating, and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan.

Allocation of ATT	Minutes
(Sum of all EOs/POs)	OJT
Enabling Checks (ECs)	OJT
Performance Checks (PCs)	OJT
Actual Training Time (ATT) Sub-total	N/A
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22	
Base Administrative Time (BAT)	N/A
Training Administrative Time (TAT)	N/A
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	N/A
Sub-total	N/A
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	N/A
For Scheduling duration total mins/400 = # days	Days
50 minutes/400 =	N/A

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex F if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO

Mentor	MS	EWK VIC	1	All
Tester	PO1	Senior Mar Tech - Mech	1	PC
Tester	PO1	Senior Mar Tech - Elect	1	PC

Critical Training Resources	No of Days
VIC Class Submarine	

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS.

a. Maximum – N/A;

b. Minimum – N/A; and

c. Optimum – N/A.

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NPTG is to be included as an INFO addressee on all loading messages.

6. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. Basic Submarine Qualification (BSQ) OJPR Harbour Watchkeeper; and

b. RQ-OS Marine Technician MITE ID 121177 NQual ALJS.

7. Qualification. Successful completion of this OJPR constitutes the primary means of attaining the Roundsperson Victoria Class Qualification (ALVD). The following MITE Course ID will be granted VIC Class Mar Tech Roundsperson - MITE Course ID 122608;

COURSE REPORTING

8. Upon completion of this OJPR, trainees shall receive a Certificate of Military Achievement (CF 289) IAW NPTORD 4500-51- Course Reports and Certification.

EVALUATION

9. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation of Instructional Programmes.

VALIDATION

10. This training will be validated IAW the guidance provided in A-P9-050-000 CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B of this document.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

11. PLARs shall be conducted in accordance with NAVORD 4500-2 – RCN PLAR Process, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

12. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED
COMMANDER NAVAL TRAINING GROUP (CNTG)

NFA (A) Approval - 10 07 2020
NTG HQ Approval - 18 10 2024
Version No 1.0
MITE Crse 122608

PO 001 Perform Submarine Rounds Forward

1. Performance. Perform Submarine Rounds Forward.

2. Conditions.

   a. Given:

      (1) Victoria Class Submarine;
      (2) PPE;
      (3) Drawings;
      (4) References;
      (5) Tools; and
      (6) Test equipment.

   b. Denied: Nil.

   c. Environment: Onboard Victoria Class Submarine, any conditions, day or night, at sea or alongside.

3. Standard. The MCC/SCC Watchkeeper shall perform submarine rounds forward, IAW refs, by:

   a. Operating the Induction System by:

      (1) Conducting SOP line-ups on the induction system;
      (2) Conducting SOP line-ups on ships ventilation system as part of pre-sea checks;
      (3) Lining up to flood upon diving;
      (4) Draining down for snorting;
      (5) Monitoring during snorting operations;
      (6) Lining-up to break the vacuum;
      (7) Securing under normal circumstances, and
      (8) IAW VCEOP's, securing under emergency circumstances.

   b. Operating the LP blow system by:(YT0130, YT0150)

      (1) Conducting SOP line-ups as part of per-sea checks in accordance with VCSOP's;
      (2) Conducting pre-start checks IAW SOPs, to include:
         (i) Alarms and trips; and
         (ii) Open/shut drains.

      (3) Lining up Ventilation Exhaust (VEX) System for:
         (i) Blowing around;
         (ii) Smoke clearance; and
         (iii) Surfacing.

(4) Securing the VEX system IAW VCSOP's.

c. Operating the AV&B system to include:

(1) Lining up the AV&B supply to the ship's whistle; and
(2) Isolating the fuel firesafe valves using AV 627 and AV240 IAW VCSOP's.

d. Operating the Voice Pipe System to prepare Voice Pipe for Operation;

e. Operating the Conning Tower Floor/Drain System by explaining the procedure for draining down the conning tower;

f. Inspecting Main Battery to include:

(1) Performing a battery dip to include:
(i) Calculating battery percentage by:
1. Taking specific gravity; and
2. Taking battery temperature readings.
(ii) Transcribing readings onto the battery log and battery state boards; and
(iii) Reporting the battery state to the OOW and the Captain.

(2) Inspecting battery sumps, to include: (KT0170)

(i) Describing actions upon finding moisture;
(ii) Operating battery sump suction as directed by the SCC1 while pumping battery sumps; and
(iii) Inspect battery cooling header pipe. (KT0140).

(3) Operating battery ventilation pneumatic (PV) valves in all modes.

g. Operating Masts and Periscopes to include:

(1) Using the local hydel:
(2) Using the TTVC; and
(3) Conducting a greasing routine on the periscopes, to include: (KT0290)

(i) Using proper grease; and
(ii) Filing a grease gun.

(4) Inspecting the drain cans for the induction and exhaust masts and explaining their purpose.

h. Operating Buoyant Wire Aerial (BWA) to include: (YT0080)

(1) Using ships drawings and VCSOP's conduct SOP Line-ups on BWA as part of Pre-sea checks;
(2) Assisting in the deployment of the BWA;

(3) Assisting in the recovery of the BWA; and
(4) Re-fresh the 'Nip' on the BWA.

i. Conducting a Set of Oncoming Engineering Rounds to include:

(1) Securing all spaces for sea;
(2) Ensuring all firefighting gear is correct;
(3) Ensuring all spaces are clean;
(4) Ensuring all bilge levels are correct;
(5) Ensuring all replenishment tank levels are correct;
(6) Ensuring all drain can levels are correct;
(7) Identifying the submarines current machinery state; and
(8) Ensuring all machinery logs are complete.

j. Operating the Surveillance System to include: (YT0210, YT0220)

(1) Monitoring Surveillance from an LSU in all modes, to include:

(i) Inhibiting/enabling a channel;
(ii) Identifying inhibited channels; and
(iii) Creating an SCC/MCC Operator Assist page.

k. Operating Trim, Bilge and Ballast System Domestic Services to include:

(1) Operating the HP Bilge and Ballast priming system cross- connection, to include:

(i) PS 610;
(ii) PS 402; and
(iii) PS 602.

(2) Operating the HP Bilge/Ballast pumps locally using starter and Field Regulator Local Indicator Panel (FRLIP); and

(3) Priming the centrifugal pumps, to include:

(i) Priming the LP Bilge Pump;
(ii) Priming the Trim Pumps; and
(iii) Priming the HP Bilge and Ballast Priming Pumps Sealing Tanks.

l. Operating Reverse Osmosis Desalinator (ROD) to include:

(1) Using ships drawings and VCSOP's conduct SOP Line-ups on ROD as part of Pre-sea checks;
(2) Conducting start/stop of the Reverse Osmosis Desalinator, by completing a set of rounds and logging in log book;
(3) Securing the ROD for domestic routines; and
(4) Testing a water sample for chlorination and pH level.

m. Operate Main/External Hydraulic Plant to include (YT0100, YT0110)

(1) Local Operation of plant;
(2) Manual operation of Unloader; and
(3) Remote Operation of plant.

4. References: VCSOP AL6 Vol 1&2, BRF 1966(12)02 Induction, BRF 1966(11)04 VEX, BRF 1966(11) LP Blow, 1966(11)05 AV&B, BR 2075 (KX) Main Battery, BR 1966(15) LP Bilge, BRF 1966(5)05 Battery ventilation controllers, BRF 1966(23) TTVC, BRF 1966(14) Main/External Hydraulics, BRF 1966(27) BWA, BRF 1966(20) Surveillance, BRF 1966(15) HP Bilge&Ballast Priming, BRF 1966(18) ROD.

5. Task Numbers: KT0140, KT0170, YT0080, YT0100, YT0110, YT0130, YT0150, YT0210, YT0220.

6. Training Limitations: Availability of a VIC Class Submarine.

PO 002 Perform Submarine Rounds Aft

1. Performance. Conduct submarine rounds Aft.

2. Conditions.

   a. Given:

      (1) Victoria Class Submarine;
      (2) PPE;
      (3) Drawings;
      (4) References;
      (5) Tools; and
      (6) Test equipment.

   b. Denied: Nil.

   c. Environment: Onboard Victoria Class Submarine, any conditions, day or night, at sea or alongside.

3. Standard. The MCC/SCC Watchkeeper shall conduct submarine rounds Aft, IAW refs, by:

   a. Inspecting Diesel Systems to include:

      (1) Completing pre-start checks on the engine;
      (2) Completing engine rounds while running by recording rounds in the engine log book;
      (3) Conducting a Kittiwake lube oil test, to include:
         (i) Lubing oil viscosity;
         (ii) Checking water-in-oil content;
         (iii) TBN number; and
         (iv) Stating the frequency that each are conducted.

      (4) Assisting the MCC1 Watchkeeper in changing over fuel from Coalescer to Centrifuge.

   b. Transferring Lubrication Oil to Equipment to include:

      (1) Using ships drawing and VCSOP's, conducting a SOP line up in the Engine Lube Oil Transfer System;
      (2) Transferring oil from Clean Lube Oil Tank to the following:
         (i) Diesel engines sumps;
         (ii) Generator pedestal bearing;
         (iii) Main motor bearings;
         (iv) Thrust block;
         (v) Centrifuge;

(vi) Engine governor; and
(vii) LP Blower.

(3) Add oil to the HPAC; and
(4) IAW VCSOP's, operating the auto greaser as MCC3 watch keeper to include the following: (YT0060)
(i) Operating Auto Greaser; and
(ii) Refilling Auto Greaser.

c. Transferring Hydraulic Oil to Hydraulic Plants to include:

(1) Using ships drawings and VCSOP's, conducting a SOP line up on the Hydraulic Transfer System; and
(2) Transferring hydraulic oil from the Hydraulic Storage tank to the Hydraulic Rep tanks using the hydraulic transfer system.

d. Operating HP Air System to include:

(1) Operating High Pressure Air system, to include:
(i) Starting-up/Shutting-down of HP air compressor;
(ii) Alarms and trips; and
(iii) Switching over air dryer IAW VCSOPs.

(2) Recharging the SCBA bottles from the SCBA charging panel; and
(3) Charging the main bottle groups from ashore.

e. Inspecting Fresh Water and Chilled Water Cooling to include:

(1) Using ship's drawings and VSCOP's, conducting SOP Lineups on Chilled Water system;
(2) Conducting SOP Lineup on Freshwater Cooling System;
(3) Starting/Stopping Chilled Water Plants; (YT0090)
(4) Conducting rounds of Chilled Water System, to include:
(i) Logging parameters;
(ii) Topping up chilled water expansion tank;
(iii) Adjusting CW flow rates; and
(iv) Operating system and pump vents.

f. Operating ship's ventilation Air Treatment Unit (ATU) plenum cross connection:

(1) Identifying and operating duct heaters in all modes;
(2) Operating ATU chilled water supply solenoid valves; and
(3) Top-up Fresh Water Cooling Expansion Tanks.

4. References: VCSOP AL6 Vol 1&2, BRF 1966(08) Diesel Generator, BRF 1966(22)05 Rudder and Hydroplanes: Autoluber, BRF 1966(14)03 Hydraulic fill and transfer, BRF1966(11)07 HPA compressor, BRF 1966(11)14 HPA Dryer, BRF 1966(13)01 Chilled

water plants, BRF 1966(12) ATU

5. Task Numbers: YT0060, YT0090.

6. Training Limitations: Availability of a VIC Class Submarine.

PO 003 Perform Submarine Rounds Alongside

1. Performance. Perform submarine rounds alongside.

2. Conditions.

   a. Given:
      (1) Victoria Class Submarine;
      (2) PPE;
      (3) Drawings;
      (4) Supervision;
      (5) References;
      (6) Tools; and
      (7) Test equipment.

   b. Denied: Nil.

   c. Environmental: Onboard Victoria Class Submarine alongside.

3. Standard. The MCC/SCC Watchkeeper shall perform submarine rounds alongside, IAW refs, by:

   a. Safety and Environment Management System, to include:

      (1) Explaining the purpose of the Victoria Class Safety and Environmental management system by explaining the meaning of Due Diligence;
      (2) Demonstrating the ability to use a Digital Volt Meter to include:
         (i) Testing leads;
         (ii) Resistance;
         (iii) AC and DC voltage; and
         (iv) Insulation resistance.

      (3) Assisting with an Electrical Lockout/Tagout IAW SEMS, to include:
         (i) Using of appropriate locking devices;
         (ii) Completing lockout forms properly;
         (iii) Updating lockout log;
         (iv) Removing lockout devices; and
         (v) Updating lockout as required.

      (4) Assisting with a Mechanical Lockout/Tagout IAW SEMS, to include:
         (i) Using of appropriate locking devices;
         (ii) Completing lockout forms properly;
         (iii) Updating lockout log;
         (iv) Removing lockout devices; and
         (v) Updating lockout as required.

b. Fueling and Transferring of POLs, to include;
   (1) Participating as a member of a fueling party (valve sentry, fueling point sentry);
   (2) Identifying the members of a submarine fueling party;
   (3) Assisting with transfer of POL's, to include:
      (i) Lube oil; and
      (ii) Hydraulic oil.

   (4) IAW Vic Class SEMS, transfer LP Bilge tank contents ashore.

c. Inspecting Battery Systems, to include:
   (1) Identifying all precautions prior to entering a battery compartment, to include:
      (i) PPE;
      (ii) Warning pipes; and
      (iii) Restrictions.

   (2) Assisting in battery maintenance, to include:
      (i) All around readings;
      (ii) Agitation test;
      (iii) Cell top cleaning;
      (iv) Inspect battery sump (KT0170)
      (v) Terminal connection check; and
      (vi) Battery electrolyte top-up.

   (3) Logging maintenance in battery maintenance log;
   (4) Acting as Hydrogen Eliminator roundsperson during a gassing charge (KT0200);
   (5) Adding water to the demineralized water tank, to include:
      (i) Testing water prior to embarking;
      (ii) Embarking/disembarking demineralized water;
      (iii) Adding water to the Battery Cooling Expansion Tank; and
      (iv) Lining up for top-up.
   (6) Using ship's drawings and VCSOP's, conducting SOP Lineups on the following systems:
      (i) Battery cooling; and
      (ii) Ventilation.

d. Inspecting Lighting Systems, to include:
   (1) Troubleshooting Ship's Service Lighting, to include:
      (i) Confirming the operation of a lighting circuit;
      (ii) Confirming power supplies to the lighting circuit;
      (iii) Replacing components of the lighting circuit; and
      (iv) Performing a function test of the lighting circuit.

   (2) Troubleshooting Navigational Lighting, to include:
      (i) Confirming the operation of all navigational lighting;
      (ii) Confirming power supplies;
      (iii) Replacing components of the navigational lighting circuit; and

(iv) Performing a function test of the navigational lighting circuit.

e. Inspecting Motor and Generators, to include:
   (1) Confirming operation, to include:
      (i) System operation;
      (ii) Starting Permissives; and
      (iii) Trips.

   (2) Confirming power under supervision at equipment using DVM and fitted indication to include:
      (i) Checking feeder panel;
      (ii) Checking input power; and
      (iii) Checking secondary power of control transfer.

   (3) Confirm status of protective devices, to include:
      (i) Controlling fuses;
      (ii) Overload protection; and
      (iii) Motor Thermal Protection Unit (MTPU).

f. Inspecting Bulk Liquid Storage, to include:
   (1) Monitoring levels of bulk liquid tanks at surveillance and at local sounding tubes, to include:
      (i) Dipping M port tank;
      (ii) Dipping M starboard tank;
      (iii) Dipping R port tank;
      (iv) Dipping R starboard tank;
      (v) Dipping port after trim tank;
      (vi) Dipping starboard after trim tank;
      (vii) Dipping port forward trim tank;
      (viii) Dipping starboard forward trim tank;
      (ix) Dipping port WRT tank;
      (x) Dipping Starboard WRT;
      (xi) Dipping port TOT; (xii)Dipping starboard TOT;
      (xii) Dipping port domestic fresh water;
      (xiii) Dipping starboard domestic fresh water;
      (xiv) Dipping clean bilge tank;
      (xv) Dipping dirty bilge tank;
      (xvi) Dipping clean oil tank;
      (xvii) Dipping dirty oil tank; and
      (xviii) Dipping distilled water tank.

   (2) Identifying which tanks have AV&B supplied blows, to include:
      (i) Locating reducers;
      (ii) Locating vent valves; and
      (iii) Checking pressures of blows.

g. Connecting Submarine to Shore Power, to include:

(1) Connecting/disconnecting AC/DC shore supply cables under supervision, to include:
    (i) Testing cable insulation with 1000V Insulation meter;
    (ii) Inspecting/Testing hull connection prior to connecting cable; and
    (iii) Connecting cable to submarine.

(2) Resetting battery switchboard links.

h. Inspecting Domestic Services, to include:
(1) Conducting procedure for:
    (i) Filling the FW system;
    (ii) Changing over tanks;
    (iii) Blowing off the fresh water system; and
    (iv) Resetting the fresh water heaters.

(2) Conducting procedure for blowing sewage tank to shore; and
(3) Operating fridge system IAW VCSOPs to include:
    (i) Checking Halocarbon Leak checks;
    (ii) Starting/stopping compressor;
    (iii) Running defrost cycle;
    (iv) Resetting HP trip; and
    (v) Logging rounds.

4. References: SEMS,VCSOP Vol 1&2, BR 2075(KX), BRF 1966(7) AC systems, BRF 1966(5)04 Battery cooling and top up system, BRF 1966(15) Trim System, BRF 1966(17)02 Domestic FW, BRF 1966(24)01 Refrigeration System

5. Task Numbers: KT0170, KT0200

6. Training Limitations: Availability of a VIC Class Submarine.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE CURRICULM CONTROL AUTHORITY

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of practical examination and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation

5. PC part one will be conducted by the submarine's senior Mar Tech Mech at the rank of PO1 or above. Part two of the PC will be conducted by the submarine's senior Mar Tech /Elect at the rank of PO1 or above. Enabling Checks are the responsibility of MCC/SCC qualified submarine crew.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of the Training PO. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and VCSOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training; and

c. Information for the TRB/PRB (if required).

8. Trainees experiencing difficulty in any area of performance are to be counselled

with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs and NPTORD 4500-54 – Training Review Board Process (TRB).

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment SOs or SOPs and NPTORD 4500-54 – Training Review Board Process (TRB). Supplemental PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their PC failure.

10. Record of trainee counselling is to be maintained by the training establishment.

TRAINING REVIEW BOARD

11. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

12. Certificate of Military Achievement (CF 289) shall be prepared and distributed IAW DAOD, 5031-9 - Course Reporting and Certificates and amplified in NPTORD 4500-51 – Course Reporting and Certificates.

13. Trainees at the minimum will be graded pass or fail.

14. Training limitations and / or incomplete training must also be included in the OJPR.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

NFA (A) Approval - 10 07 2020
NTG HQ Approval - 18 10 2024
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MITE Crse 122608

PO 001 Perform Submarine Rounds Forward

EO 001.01 Operate the Induction System

1. Performance. Operate the Induction System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Onboard Victoria Class Submarine.

3. Standard. The trainee shall operate the induction system, IAW refs, by:

   a. Conducting SOP line-ups on the induction system;
   b. Conducting SOP line-ups on ships ventilation system as part of pre-sea checks;
   c. Lining up to flood upon diving;
   d. Draining down for snorting;
   e. Monitoring during snorting operations;
   f. Lining-up to break the vacuum;
   g. Securing under normal circumstances, and
   h. IAW VCSOP's, securing under emergency circumstances.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard will be assessed in PC 001.

8. References: VCSOP AL6 Vol 1&2, BRF 1966(12)02 Induction, BRF 1966(11)04 VEX, BRF 1966(11) LP Blow, 1966(11)05 AV&B, BR 2075 (KX) Main Battery, BRF1966(15) LP Bilge, BRF 1966(5)05 Battery ventilation controllers, BRF 1966(23) TTVC, BRF 1966(14) Main/External Hydraulics, BRF 1966(27) BWA, BRF 1966(20) Surveillance, BRF 1966(15) HP Bilge&Ballast Priming, BRF 1966(18) ROD

9. Remarks: Nil

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.01

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.02 Operate the LP Blow System

1. Performance. Operate the LP Blow System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Onboard Victoria Class Submarine.

3. Standard. The trainee shall operate the LP Blow System, IAW refs, by:

   a. Conducting SOP line-ups as part of per-sea checks in accordance with VCSOP's;
   b. Conducting pre-start checks IAW SOPs, to include:
      (1) Alarms and trips; and
      (2) Open/shut drains.

   c. Lining up Ventilation Exhaust (VEX) System for:
      (1) Blowing around;
      (2) Smoke clearance; and
      (3) Surfacing.

   d. Securing the VEX system IAW VCSOP's.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details.

   a. EO 001.02 will be tested in EC 001.02 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References. VCSOP AL6 Vol 1&2, BRF 1966(12)02 Induction, BRF 1966(11)04 VEX, BRF 1966(11) LP Blow, 1966(11)05 AV&B, BR 2075 (KX) Main Battery, BRF1966(15) LP Bilge, BRF 1966(5)05 Battery ventilation controllers, BRF 1966(23) TTVC, BRF 1966(14) Main/External Hydraulics, BRF 1966(27) BWA, BRF 1966(20) Surveillance, BRF 1966(15) HP Bilge&Ballast Priming, BRF 1966(18) ROD.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.02

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.03 Operate the AV&B System

1. Performance. Operate the AV&B System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil

   c. Environment: Victoria Class Submarine

3. Standard. The trainee shall operate the AV&B system, IAW refs, by:

   a. Lining up the AV&B supply to the ship's whistle; and
   b. Isolating the fuel firesafe valves using AV 627 and AV240 IAW VCSOP's.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.03 will be tested in EC 001.03 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References. VCSOP AL6 Vol 1&2, BRF1966(11)05 AV&B.

9. Remarks: Nil.

10. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.03

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.04 Operate the Voice Pipe System

1. Performance. Operate the Voice Pipe System

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environment: Victoria Class Submarine.

3. Standard. The trainee shall operate the voice pipe system, IAW refs, by preparing Voice Pipe for Operation.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.04 will be tested in EC 001.04 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 2, BRF 1966(12)02 Induction, 1966(11)05 AV&B.

9. Remarks: Nil.

10. Teaching Points.

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.04

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.05 Operate the Conning Tower Flood/Drain System

1. Performance. Operate the Conning Tower Flood/Drain System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall operate the conning tower flood/drain, IAW refs, by explaining the procedure for draining down the conning tower.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.05 will be tested in EC 001.05 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 2,1966(11)05 AV&B, BRF 1966(14) Main/External Hydraulics.

9. Remarks: Nil.

10. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.05

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.06 Inspect Main Battery

1. Performance. Inspect Main Battery.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall inspect the Main Battery, IAW refs, by:

   a. Performing a battery dip to include:

      (1) Calculating battery percentage by:
         (i) Taking specific gravity; and
         (ii) Taking battery temperature readings.

      (2) Transcribing readings onto the battery log and battery state boards; and

      (3) Reporting the battery state to the OOW and the Captain,

   b. Inspecting battery sumps, to include:

      (1) Describing actions upon finding moisture; and
      (2) Operating battery sump suction as directed by the SCC1 while pumping battery sumps
      (3) Inspect battery cooling header pipes.

   c. Operating battery ventilation pneumatic (PV) valves in all modes.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;

b. Self-study; and
c. OJT.

6. Training Aids: NA

7. Test Details :
   a. EO 001.06 will be tested in EC 001.06 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References VCSOP AL6 Vol 1&2, BR 2075 (KX) Main Battery.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.06

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.07 Operating Masts and Periscopes

1. Performance. Operating Masts and Periscopes.

2. Conditions

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall operate masts and periscopes, IAW refs, by:

   a. Operating masts and periscopes, to include:
      (1) Using the local hydel; and
      (2) Using the TTVC;

   b. Conducting a greasing routine on the periscopes, to include:
      (1) Using proper grease; and
      (2) Filing a grease gun; and

   c. Inspecting the drain cans for the induction and exhaust masts and explaining their purpose.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.07 will be tested in EC 001.07 by demonstrating the standard to a qualified MCC/SCC

Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References. VCSOP AL6 Vol 1&2, BRF 1966(12)02 Induction, BRF 1966(11)04 VEX, BRF 1966(11) LP Blow, BRF 1966(14) Main/External Hydraulics.

9. Remarks: Nil.

10. Teaching Points.

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.07

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.08 Operating Buoyant Wire Aerial (BWA)

1. Performance. Operating Buoyant Wire Aerial (BWA).

2. Conditions

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environment: Victoria Class Submarine.

3. Standard. The trainee shall operate the Buoyant Wire Aerial, IAW refs, by:

   a. Using ships drawings and VCSOP's conduct SOP Line-ups on BWA as part of Pre-sea checks;
   b. Assisting in the deployment of the BWA;
   c. Assisting in the recovery of the BWA; and
   d. Re-fresh the 'Nip' on the BWA.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.08 will be tested in EC 001.08 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC

8. References. BRF 1966(27) BWA.
9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.08

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.09 Conducting a Set of Oncoming Rounds

1. Performance. Conducting a Set of Oncoming Rounds.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall perform a set of oncoming rounds, IAW refs, by:

   a. Securing all spaces for sea;
   b. Ensuring all firefighting gear is correct;
   c. Ensuring all spaces are clean;
   d. Ensuring all bilge levels are correct;
   e. Ensuring all replenishment tank levels are correct;
   f. Ensuring all drain can levels are correct;
   g. Identifying the submarines current machinery state; and
   h. Ensuring all machinery logs are complete.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.09 will be tested in EC 001.09 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC

001.

8. References VCSOP AL6 Vol 1&2, BRF 1966(12)02 Induction, BRF 1966(11)04 VEX, BRF 1966(11) LP Blow, 1966(11)05 AV&B, BR 2075 (KX) Main Battery, BRF 1966(15) LP Bilge, BRF 1966(5)05 Battery ventilation controllers, BRF 1966(23) TTVC, BRF 1966(14) Main/External Hydraulics, BRF 1966(27) BWA, BRF 1966(20) Surveillance, BRF 1966(15) HP Bilge & Ballast Priming, BRF 1966(18) ROD.

9. Remarks: Nil.

10. Teaching Points.

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.09

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.10 Operating the Surveillance System

1. Performance. Operating the Surveillance System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee operate the surveillance system, IAW refs, by:

   a. Monitoring Surveillance from an LSU in all modes, to include:
      (1) Inhibiting/enabling a channel;
      (2) Identifying inhibited channels; and
      (3) Creating an SCC/MCC Operator Assist page.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.10 will be tested in EC 001.10 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 1&2, BRF 1966(20) Surveillance.

9. Remarks: Nil.

10. Teaching Points.

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.10

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.11 Operate Trim, Bilge and Ballast System Domestic Services

1. Performance. Operate the Trim, Bilge and Ballast System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall operate the trim, bilge and ballast system, IAW refs, by:

   a. Operating the HP Bilge and Ballast priming system cross-connection, to include:
      (1) PS 610;
      (2) PS 402; and
      (3) PS 602;

   b. Operating the HP Bilge/Ballast pumps locally using starter and Field Regulator Local Indicator Panel (FRLIP); and

   c. Priming the centrifugal pumps, to include:
      (1) Priming the LP Bilge Pump;
      (2) Priming the Trim Pumps; and
      (3) Priming the HP Bilge and Ballast Priming Pumps Sealing Tanks.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 001.11 will be tested in EC 001.11 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 1&2, BRF 1966(15) LP Bilge, BRF 1966(15) HP Bilge&Ballast Priming, BRF 1966(18) ROD.

9. Remarks: Nil.

10. Teaching Points.

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.11

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 001.12 Operating Reverse Osmosis Desalinator (ROD)

1. Performance. Operating Reverse Osmosis Desalinator (ROD).

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall operate the Reverse Osmosis Desalinator (ROD), IAW refs, by:

   a. Using ships drawings and VCSOP's conduct SOP Line-ups on ROD as part of Pre-sea checks;
   b. Conducting start/stop of the Reverse Osmosis Desalinator, by completing a set of rounds and logging in log book;
   c. Securing the ROD for domestic routines; and
   d. Testing a water sample for chlorination and pH level.

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.12 will be tested in EC 001.12 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 1&2, BRF 1966(18) ROD.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 001.12

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

PERFORMANCE CHECK – PO 001

1. Trainee will be provided feedback through PO 001. To achieve the intended assessment of PC 001, the trainee must achieve a qualifying signature in every signature block associated to PO 001 and successfully complete the walk-through in the OJT package found in Annex D.

PO 002 Perform Submarine Rounds AFT

EO 002.01 Inspect Diesel Systems

1. Performance. Inspect Diesel Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environment: Victoria Class Submarine.

3. Standard. The trainee shall inspect diesel systems, by:

   a. Completing pre-start checks on the engine;
   b. Completing engine rounds while running by recording rounds in the engine log book;
   c. Conducting a Kittiwake lube oil test, to include:

      (1) Lubing oil viscosity;
      (2) Checking water-in-oil content;
      (3) TBN number; and
      (4) Stating the frequency that each are conducted; and

   d. Assisting the MCC1 Watchkeeper in changing over fuel from Coalescer to Centrifuge.

4. Time Required

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

a. Demonstration - performance;
b. Self-study; and
c. OJT.

6. Training Aids: Victoria Class Submarine

7. Test Details:

a. EO 002.01 will be tested in EC 002.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 1&2, BRF 1966(8) 01-19 DIESEL ENGINES, KITTIWAKE MANUAL.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 002.01

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 002.02 Transfer Lubrication Oil to Equipment

1. Performance. Transfer Lubrication Oil to Equipment.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall transfer lubrication oil to equipment, by:

   a. Using ships drawing and VCSOP's, conducting a SOP line up in the Engine Lube Oil Transfer System;
   b. Transferring oil from Clean Lube Oil Tank to the following:

      (1) Diesel engines sumps;
      (2) Generator pedestal bearing;
      (3) Main motor bearings;
      (4) Thrust block;
      (5) Centrifuge;
      (6) Engine governor; and
      (7) LP Blower.

   c. Add oil to the HPAC;
   d. IAW VCSOP's, operating the auto greaser as MCC3 watch keeper to include the following:

      (1) Operating Auto Greaser; and
      (2) Refilling Auto Greaser.

4. Time Required

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

a. Demonstration - performance;
b. Self-study; and
c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 002.02 will be tested in EC 002.02 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 2, BRF 1966(08) Diesel Generator, BRF1966(11)07 HPA compressor, BRF 1966(11)14 HPA Dryer, BRF 1966(13)01 Chilled water plants, SEMS

9. Remarks: Nil

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 002.02

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 002.03 Transferring Hydraulic Oil to Hydraulic Plants

1. Performance. Transferring Hydraulic Oil to Hydraulic Plants.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall transfer hydraulic oil to the hydraulic plants, IAW refs, by:

   a. Using ships drawings and VCSOP's, conducting a SOP line up on the Hydraulic Transfer System; and
   b. Transferring hydraulic oil from the Hydraulic Storage tank to the Hydraulic Rep tanks using the hydraulic transfer system.

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 002.03 will be tested in EC 002.03 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 2, BRF 1966(14)03 Hydraulic fill and transfer, SEMS.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 002.03

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 002.04 Operating HP Air System

1. Performance. Operating HP Air System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds AFT, IAW refs, by:

   a. Operating High Pressure Air system, to include:

      (1) Starting-up/Shutting-down of HP air compressor;
      (2) Alarms and trips; and
      (3) Switching over air dryer IAW VCSOPs;

   b. Recharging the SCBA bottles from the SCBA charging panel; and
   c. Charging the main bottle groups from ashore;

4. Time Required.

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 002.04 will be tested in EC 002.04 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 2, BRF1966(11)07 HPA compressor.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 002.04

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 002.05 Inspecting Fresh Water and Chilled Water Cooling

1. Performance. Inspecting Fresh Water and Chilled Water Cooling. (May have T000 at the end)

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds AFT, IAW refs, by:

   a. Using ship's drawings and VSCOP's, conducting SOP Lineups on Chilled Water system;
   b. Conducting SOP Lineup on Freshwater Cooling System;
   c. Starting/Stopping Chilled Water Plants;
   d. Conducting rounds of Chilled Water System, to include:

      (1) Logging parameters;
      (2) Topping up chilled water expansion tank;
      (3) Adjusting CW flow rates; and
      (4) Operating system and pump vents;

   e. Operating ship's ventilation Air Treatment Unit (ATU) plenum cross connection;
   f. Identifying and operating duct heaters in all modes;
   g. Operating ATU chilled water supply solenoid valves; and
   h. Top-up Fresh Water Cooling Expansion Tanks.

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and

c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 002.05 will be tested in EC 002.05 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP AL6 Vol 2, BRF 1966(13)01 Chilled water plants, BRF 1966(17)01 Ships Freshwater cooling.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 002.05

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

PERFORMANCE CHECK – PO 002

1. Trainee will be provided feedback through PO 002. To achieve the intended assessment of PC 002, the trainee must achieve a qualifying signature in every signature block associated to PO 002 and successfully complete the walk-through in the OJT package found in Annex D.

PO 003 Perform Submarine Rounds Alongside

EO 003.01 Safety and Environment Management System

1. Performance. Safety and Environment Management System. (May have T000 at the end).

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Explaining the purpose of the Victoria Class Safety and Environmental management system by explaining the meaning of Due Diligence;
   b. Demonstrating the ability to use a Digital Volt Meter to include:

      (1) Testing leads;
      (2) Resistance;
      (3) AC and DC voltage; and
      (4) Insulation resistance;

   c. Assisting with an Electrical Lockout/Tagout IAW SEMS, to include:

      (1) Using of appropriate locking devices;
      (2) Completing lockout forms properly;
      (3) Updating lockout log;
      (4) Removing lockout devices; and
      (5) Updating lockout as required;

   d. Assisting with an Electrical Lockout/Tagout IAW SEMS, to include:

      (1) Using of appropriate locking devices;
      (2) Completing lockout forms properly;
      (3) Updating lockout log;
      (4) Removing lockout devices; and

(5) Updating lockout as required.

4. Time Required:

a. EO - OJT; and
b. EC – OJT.

5. Instructional Methods:

a. Demonstration - performance;
b. Self-study; and
c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 003.01 will be tested in EC 003.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.01

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.02 Fueling and Transferring of POLs

1. Performance. Fueling and Transferring of POLs. (May have T000 at the end).

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Participating as a member of a fueling party (valve sentry, fueling point sentry);
   b. Identifying the members of a submarine fueling party;
   c. Assisting with transfer of POL's, to include:

      (1) Lube oil; and
      (2) Hydraulic oil.

   d. IAW Vic Class SEMS, transfer LP Bilge tank contents ashore.

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 003.02 will be tested in EC 003.02 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP Vol 2, SEMS, BRF 1966 (08)04 Diesel Engines: Fuel Oil and Compensation system.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.02

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.03 Inspecting Battery Systems

1. Performance. Inspecting Battery Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Identifying all precautions prior to entering a battery compartment, to include:

      (1) PPE;
      (2) Warning pipes; and
      (3) Restrictions.

   b. Assisting in battery maintenance, to include:

      (1) All around readings;
      (2) Agitation test;
      (3) Cell top cleaning;
      (4) Terminal connection check; and
      (5) Battery electrolyte top-up.

   c. Logging maintenance in battery maintenance log;
   d. Acting as Hydrogen Eliminator roundsperson during a gassing charge;
   e. Adding water to the demineralized water tank, to include:

(1) Testing water prior to embarking;
(2) Embarking/disembarking demineralized water;
(3) Adding water to the Battery Cooling Expansion Tank; and
(4) Lining up for top-up.

f. Using ship's drawings and VCSOP's, conducting SOP Lineups on the following systems:

(1) Battery cooling; and
(2) Ventilation.

4. Time Required:

a. EO - OJT; and
b. EC – OJT.

5. Instructional Methods:

a. Demonstration - performance;
b. Self-study; and
c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: SEMS, VCSOP Vol 1&2, BRF 1966(5) Battery cooling, VCSOP 1966(5)02 Battery ventilation.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to

							complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.03

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.04 Inspecting Lighting Systems

1. Performance. Inspecting Lighting Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Troubleshooting Ship's Service Lighting, to include:

      (1) Confirming the operation of a lighting circuit;
      (2) Confirming power supplies to the lighting circuit;
      (3) Replacing components of the lighting circuit; and
      (4) Performing a function test of the lighting circuit.

   b. Troubleshooting Navigational Lighting, to include:

      (1) Confirming the operation of all navigational lighting;
      (2) Confirming power supplies;
      (3) Replacing components of the navigational lighting circuit; and
      (4) Performing a function test of the navigational lighting circuit.

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and

c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP Vol 2, BRF 1966(7) AC Systems, BRF 1966(30)01 Navigation external lights.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.04

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.05 Inspecting Motor and Generators

1. Performance. Inspecting Motor and Generators.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Confirming operation, to include:

      (1) System operation;
      (2) Starting Permissives; and
      (3) Trips.

   b. Confirming power under supervision at equipment using DVM and fitted indication to include:

      (1) Checking feeder panel;
      (2) Checking input power; and
      (3) Checking secondary power of control transfer.

   c. Confirm status of protective devices, to include:

      (1) Controlling fuses;
      (2) Overload protection; and
      (3) Motor Thermal Protection Unit (MTPU).

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

   a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: BRF 1966(7) AC System and equipment.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.05

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.06 Inspecting *Bulk Liquid Storage*

1. Performance. Inspecting Bulk Liquid Storage.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Monitoring levels of bulk liquid tanks at surveillance and at local sounding tubes, to include:

      (1) Dipping M port tank;
      (2) Dipping M starboard tank;
      (3) Dipping R port tank;
      (4) Dipping R starboard tank;
      (5) Dipping port after trim tank;
      (6) Dipping starboard after trim tank;
      (7) Dipping port forward trim tank;
      (8) Dipping starboard forward trim tank;
      (9) Dipping port WRT tank;
      (10) Dipping Starboard WRT;
      (11) Dipping port TOT;
      (12) Dipping starboard TOT;
      (13) Dipping port domestic fresh water;
      (14) Dipping starboard domestic fresh water;

(15) Dipping clean bilge tank;
(16) Dipping dirty bilge tank;
(17) Dipping clean oil tank;
(18) Dipping dirty oil tank; and
(19) Dipping distilled water tank.

b. Identifying which tanks have AV&B supplied blows, to include:

(1) Locating reducers;
(2) Locating vent valves; and
(3) Checking pressures of blows.

4. Time Required:

a. EO - OJT; and
b. EC – OJT.

5. Instructional Methods:

a. Demonstration - performance;
b. Self-study; and
c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP Vol 2, BRF 1966(15)01 Trim system, BRF 1966(11)05 Auxiliary vent and blow.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
		OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's

							responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points,
							demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.06

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.07 Connecting Submarine to Shore Power

1. Performance. Connecting Submarine to Shore Power.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Connecting/disconnecting AC/DC shore supply cables under supervision, to include:

      (1) Testing cable insulation with 1000V Insulation meter;
      (2) Inspecting/Testing hull connection prior to connecting cable; and
      (3) Connecting cable to submarine; and

   b. Resetting battery switchboard links.

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

   a. Demonstration - performance;
   b. Self-study; and
   c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: VCSOP Vol 1&2.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.07

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

EO 003.08 Inspecting Domestic Services

1. Performance. Inspecting Domestic Services.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Victoria Class Submarine.

3. Standard. The trainee shall Perform Submarine Rounds Alongside, IAW refs, by:

   a. Conducting procedure for:

      (1) Filling the FW system;
      (2) Changing over tanks;
      (3) Blowing off the fresh water system; and
      (4) Resetting the fresh water heaters.

   b. Conducting procedure for blowing sewage tank to shore; and
   c. Operating fridge system IAW VCSOPs to include:

      (1) Checking Halocarbon Leak checks;
      (2) Starting/stopping compressor;
      (3) Running defrost cycle;
      (4) Resetting HP trip; and
      (5) Logging rounds.

4. Time Required:

   a. EO - OJT; and
   b. EC – OJT.

5. Instructional Methods:

a. Demonstration - performance;
b. Self-study; and
c. OJT.

6. Training Aids: Victoria Class Submarine.

7. Test Details:

a. EO 001.01 will be tested in EC 001.01 by demonstrating the standard to a qualified MCC/SCC Watchkeeper and having the corresponding OJPR signed. Standard can be assessed in PC 001.

8. References: SEMS, VCSOP Vol 2, BRF 1966(17_02 Domestic FW system, BRF 1966(25)07 Domestic Automatic refrigerators.

9. Remarks: Nil.

10. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
N/A	N/A	OJT	OJT	OJT	F2F; Self-Study		While it is the trainee's responsibility to complete the OJT and learn material, Submarine staff shall explain teaching points, demonstrate performance, and supervise practice.

Instructional technique – T=Theory, D= Demonstration, and P=Practical

Enabling check - EO 003.08

1. Trainees will be provided feedback throughout this EO formatively testing the theory content. The practical content within this EO will be assessed through the signature block found in Annex D.

PERFORMANCE CHECK – PO 003

1. Trainee will be provided feedback through PO 003. To achieve the intended assessment of PC 003, the trainee must achieve a qualifying signature in every signature block associated to PO 003 and successfully complete the walk-through in the OJT package found in Annex D.

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Code	Document Number	Publication
A1		VCEOP's
A2		NAVORD's
A3		NEM
A4		VCSSO's
A5		VCSOP's Vol. 1 &2
A6		Victoria Class SEMS
A7	C-23-VIC-000	Quality Assurance for First Level Systems and Areas
A8	MARPACORD G-108	MARPAC Lockout/Tagout Program
A9	MARLANTORD 66-3	MARLANT Lockout/Tagout Program
A10	C-69-820-000/MB-008	SUBWOPS Vol. 3
B1	BR 1966(2)	Firefighting Systems & Equipment
B2	BR 1966(3)	Escape and Salvage Systems and Equipment
B3	BR 1966(4)	VPDC Systems & Equipment – Index
B4	BR 1966(5)	Main Battery Systems & Equipment – Index
B5	BR 1966(6)	24 VDC Systems & Equipment – Index
B6	BR 1966(7)	AC Systems & Equipment – Index
B7	BR 1966(8)	Diesel Engines, Installation, Systems, & Equipment – Index
B8	BR 1966(9)	Prop: Electrical Systems & Equipment – Index
B9	BR 1966(10)	Prop: Mechanical Systems & Equipment – Index
B10	BR 1966(11)	Compressed Air & Nitrogen Systems & Equipment – Index
B11	BR 1966(12)	Ventilation, Air Conditioning & Snort Induction – Index
B12	BR 1966(13)	Chilled Water Systems & Equipment – Index
B13	BR 1966(14)	Ships Hydraulic Systems & Equipment – Index
B14	BR 1966(15)	Trim, Bilge & Ballast Systems
B15	BR 1966(16)	Ships SW System & Equipment
B16	BR 1966(17)	Ships & Domestic FW System & Equipment
B17	BR 1966(18)	Desalination Plant
B18	BR 1966(20)	Surveillance System and Equipment
B19	BR 1966(21)	Control Consoles – Index
B20	BR 1966(22)	Rudder & Hydroplanes, Control System & Equip – Index
B21	BR 1966(23)	Masts & Associated Equipment – Index
B22	BR 1966(24)	Ship's Stores Refrigeration

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Code	Document Number	Publication
B23	BR 1966(25)	Domestic Equipment
B24	BR 1966(30)	Lighting Systems & Equipment – Index
B25	BR 2075(K)	Submarine Main Batteries
B26	BR 3623(01)	TAB's – Ships Systems
B27	BR 3623(02)	TAB's – Propulsion
B28	BR 3623(03)	TAB's – Tactical Weapons Systems
B29	BR 3623(04)	TAB's – Electrical Systems
B30	BR 4500(526)	Static Frequency Changer, MK38SFC
B31	BR 4500(529)	Static Frequency Changer, MK36SFC
B32	BR 4500(701)	Batteries, Primary & Secondary
B33	BR 4569	Pressure & Differential Pressure Transducers & Associated Equipment
B34	BR 6528(301)	Paxman Valenta Diesel Engine 16 RPA 200z
B35	BR 6528(303)	Valenta Engines Dockyard Overhaul/Repair
B36	BR 8223	Breathing Apparatus
C1		Propulsion Power Statement
C2		Composite Drawings
C3		Autopilot Manual
C4		CSS Manual
C5	DRMIS-SIGRD.mil.ca/index.eng.asp	DRMIS Home Page

ANNEX B – TASK LIST

Task No	Task Statement	Trg / No Trg	PO
KT0030	Assemble / disassemble auto lubrication system components (sub)	Trg	
KT0060	Assemble / disassemble chilled water system components (sub)	Trg	
KT0140	Inspect battery cooling header pipe (sub)	Trg	
KT0170	Inspect battery sump (sub)	Trg	
KT0200	Inspect hydrogen eliminators (sub)	Trg	
KT0250	Inspect visually, auxiliary vent and blow system (subs)	No Trg	
KT0290	Lubricate periscope (Sub)	Trg	
KT0470	Test alarm relay outputs	No Trg	
YT0060	Operate auto lubrication system	Trg	
YT0080	Operate Buoyant Wire Arial transfer system (subs)	Trg	
YT0090	Operate chilled water system (subs)	Trg	
YT0100	Operate hydraulic plant locally (subs)	Trg	
YT0110	Operate hydraulic Plant remotely (subs)	Trg	
YT0120	Operate main ballast tank high pressure blow system remotely (Sub)	No Trg	
YT0130	Operate main ballast tank low pressure blow system remotely (Sub)	Trg	
YT0140	Operate main ballast tank high pressure blow system locally (Sub)	Trg	
YT0150	Operate main ballast tank low pressure blow system locally (Sub)	Trg	
YT0180	Operate ship's ventilation systems locally (subs)	Trg	
YT0190	Operate ship's ventilation systems remotely (subs)	Trg	
YT0200	Operate submarine battery ventilation system locally	Trg	
YT0210	Operate surveillance system locally (sub)	Trg	
YT0220	Operate surveillance system remotely (Sub)	Trg	

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 28 Jan 2019 – 01 Feb 2019

Location: CFB Halifax, Building S-17

Board Members:

Surname	Initials	Rank	MOSID	Unit
Byrnes (Chair)	R.S.	CPO2	00379-6	Sea Training Atlantic
Guthrie (TDO)	N.A.	SLt	00211	NTDC(A)
Hamilton (SME)	M.A.	PO2	00379-5	HMCS Windsor
Mohr (SME)	G.D.	PO1	00379-1	HMCS Corner Brook
Jacobson (SME)	I.R.	LS	00379-2	HMCS Windsor
Linzmeyer (SME)	K.M.	PO2	00379-5	NFSA
Cyr (SME)	C.J.	PO2	00379-3	NFSA
Thompson (SME)	J.L.	CPO2	00379-1	Sea Training Atlantic

Board Objective:

To produce the Marine Technician (Mar Tech – 00379) Basic Maintainer Roundsperson qualification for the Victoria Class Submarines.

28 Jan 2019

Board convened and discussed the current situation concerning the amalgamation of marine systems trades (H Tech, E Tech, and Mar Eng). One common RQ-Marine Technician – OS (ALJS) was created and a submarine class package will need to be completed to complement this training.

Target Population Analysis indicated that students will be Marine Technicians who have completed the RQ-OS (ALJS) and Basic Submarine Qualification (BSQ).

The two legacy trades on-board a submarine are E Tech and Mar Eng. Both the E Tech and Mar Eng previously completed the AMOC to be qualified to stand as SCC2 and MCC3. The E Techs would also complete an E Tech Jr Experience Package / OJPR. The board discussed combining the two packages with little change since the job expectations

on the submarine have not changed. New Mar Techs require the E Tech skills as they will now be performing the E Tech duties.

Board examined the new RQ-OS (ALJS) QSP and class packages of the other platforms.

Board completed a train/no train analysis of the specific submarine task on the Master Task List (MTL) in the Marine Technician Occupational Specification (A-PD-055-002/PP-002) dated 2019-03-23. Only tasks under the Submarine duty areas (duty areas K and Y) were reviewed as, after a quick search, all other tasks appeared to be covered by the RQ-OS QSP.

It was decided that the board would review the signatures required in the legacy AMOC package to decide what items should remain or be removed;

- Turn over brief was added, as a SCC2/ forward roundsperson would do this task.
- Under SCC2 board felt that the roundsperson should do more than monitor the LSU. Some operational tasks were added.
- Due to few crew members being trained in conduction fresh water test, board members added this to the package to train more people to do the task to allow the Submarine to even out the work load.
- "Explaining the procedures for flooding the conning tower" was removed. Board agreed that it is not necessary at this level of training. They would not be responsible for this.
- The restarting of Chilled Water Plants was added. This would be a basic start as if the plants were running and shut down. Not a full chiller set-up and start.

All system sketches were removed. Board had some discussion on how techs would learn the systems. Board discussed having the trainees answer questions about the systems and possibly explaining the system using a given system sketch. TDO brought up that answering questions about a sketch is not a task that would normally done to perform their roundsperson duties and also raised concerns about standardizing the questions. Board members found in the HFX roundsperson package under the Annex E instructions it states, "A series of questions will be asked from each EO to satisfy the knowledge of the roundsperson before granting qualification." This was taken to mean that in the HFX package they will be asking questions, therefore there shouldn't be any reason we couldn't in this package.

There were also discussions on having a board and residential phase to enhance system knowledge. It was decided the board will be visited later after seeking guidance from TM and TDO working on OS-RQ and other class packages.

Some members of the board felt that the BSQ would cover any items that would need to be covered in a residential phase, however it was brought up that BSQ may be changing. No decision was made on a residential/classroom phase.

The board reviewed the E Tech OJPR package (Jr. Experience Package). With minor changes, most components of the package were added to new roundsperson package with the exception of the following;

- Repair navigational lighting was removed, as this would be at the USQ AB level.
- Troubleshooting and repairing ships explosion proof fixtures was removed. Done at USQ AB level.
- Troubleshoot, remove, repair, replace motor controller removed. Done at USQ AB level.

The board adjourned for the day and decided to meet a day later. This would give time for members to reach out for further direction and for the tasks to be compiled together in the new QSP template.

30 Jan 2019

Board reconvened and new OJPR package (sign off list) was reviewed. Minor changes made after board agreement.

-Vacuum test added as roundsperson may be required to reset.

-Shutoff for attack was discussed but not added.

The need for a classroom component was discussed. The HFX class package contains a classroom component. The old Submarine AMOC had no class time and the board felt that the BSQ would cover material required to commence the roundsperson OJPR. It was decided that this QSP would be entirely OJT. Theory required would be covered on the RQ-OS and BSQ courses. BSQ Harbour Watchkeeper was decided to be a pre-requisite as it would be a priority for the boat and give experience on the equipment that will be beneficial to completing this OJPR.

The HFX and KIN class packages do not have a formal PC. PC is granted after getting EC signatures. Board not comfortable with this set up. TDO stated that generally according to CFITES POs should have a PC. Qualification Boards are still being used in the fleet and after contact with NTDC(P) no conflict was found with using a Board. A walk-through was also considered. It would be easier to implement and more aligned with the roundspersons job. TDO will look into any policies concerning boards and walk-throughs. No policy was found that stated walk-throughs could be used in RCN assessments.

Board adjourned for the day to allow for updates to be made to checklist and writing of the QSP.

1 Feb 2019

Board reviewed minutes from previous days.

It was decided that a walk-though would be used as the final assessment and treated as a PC. This is more realistic than a board and more practical than multiple PC on each small task contained in the OJPR. TDO suggested that there should be three walk-throughs, Forward Rounds, After Rounds, and Alongside Rounds as this most reflects the job. Board discussed that this would not be practical as some equipment forward and aft is electrical and some mechanical. Two assessors would be required. It was decided that the walk-throughs would be divided into Electrical and Mechanical as this would be the most effective and efficient way to conduct them.

West Coast TDOs where contacted about format of the QSP. East Coast uses the FireFly format different than West. It was advised that we product the QSP using the West Coast template as they control the other Mar Tech QSPs. This would maintain consistency in the documentation for the Mar Techs.

Board members where given action items and the board was adjourned.

ANNEX D – ON JOB PERFORMANCE RECORD (OJPR)

MITE - 122608
(Certificate of Qualification)

Name and Initials Rank SN

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. Recording of the individual's progress shall be entered in Monitor Mass under the Training Tracker. The NFS for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective to the required standard, the Mar Tech Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and enter progress in the Monitor Mass Training Tracker.

3. If it is not possible to conduct OJPR for some POs due to equipment not fitted or not issued (i.e. Work Period), the Mar Tech Training PO is directed to contact NTDC(A) Submarine Division for guidance. Extension of training time might be required.

4. Review OJPR sign off list.

5. Performance Checks require the trainee to demonstrate the listed competencies as described in chapter 4. A series of questions will be asked from each EO to satisfy the knowledge of the roundsperson before granting qualification. The trainee will have three (3) months from completion of the Harbour Watchkeeper component of the Basic Submarine Qualification (BSQ) OJPR to complete Performance Checks. Failure to complete the OJPR within three month period may result in an extension or Training Review Board (TRB) IAW NPTORD 4500-54.

6. All EOs shall be performed with safety as the upmost importance and IAW VCSOPs, VCEOPs, EOTIs, VCSSOs, Vic Class SEMS, NAMMs, NEM, and COTM.

7. Upon posting or transfer, the losing unit will ensure the following:

a. The trainee's OJPR is reviewed to ensure that all of the objectives which have been successfully completed are credited;
b. A duplicate copy of the original should be made and entered into the member's PERs file for safe keeping; and
c. The master copy OJPR is forwarded with the member to the new unit.

8. The gaining unit shall review both the master and duplicate copies of the OJPR package to become familiar with:

a. Percentage of objectives completed;
b. Remarks; and
c. OJT time elapsed.

9. The Unit Training Chief will also send an email with the members information (SN, Rank, Name, Trade, OJPR, and completion date) to the following personnel:

a. +NFS(A) MITE@NFS(A)@Halifax; and

b. Completion message IAW NPTORD 4500-55, Annex A.

PROGESS REVIEW

10. Progress review shall be carried out monthly by the Divisional Officer or MSEC.

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

MSEC (Print and sign) Date

CERTIFICATION OF QUALIFICATION

I certify that_____has completed training to the standards stated in the Roundsperson on a VIC class submarine Class Package, except as noted. The member is qualified to carry out the duties of roundsperson on a VIC Class Submarine and is ALVD qualified.

_____

Divisional Officer (Print and Sign) Date

ON-JOB PERFORMANCE RECORD SHEET

11. The remarks column in the OJPR shall be used for the following information only:

a. The unit in which the training and assessment were carried out; and

b. Any unusual circumstances which preclude compliance with an objective.

Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
PO 001 Perform Submarine Rounds Forward				
EO 001.01 Operate the Induction System				
001.01.001	Conduct SOP Line-up on induction system	MCC/SCC WKPR		
001.01.002	Conduct SOP Line-up on ship's ventilation system	MCC/SCC WKPR		
001.01.003	Line up induction system to flood upon diving	MCC/SCC WKPR		
001.01.004	Drain down induction system for snorting	MCC/SCC WKPR		
001.01.005	Monitor induction system during snorting operations	MCC/SCC WKPR		
001.01.006	Secure induction system under normal circumstances	MCC/SCC WKPR		
001.01.007	Line-up induction system in order to break the vaccuum	MCC/SCC WKPR		
001.01.008	Secure the induction system in emergency IAW VCEOP's	MCC/SCC WKPR		
EO 001.02 Operate the LP Blow System				
001.02.001	Conduct SOP Line-up on LP Blow system	MCC/SCC WKPR		
001.02.002	Line up for a vacuum test IAW VCSOP's	Sr Mar Tech Mech		
001.02.003	Conduct pre-start checks of the LP blower	MCC/SCC WKPR		
001.02.004	Line-up Ventilation Exhaust (VEX) system to blow around	MCC/SCC WKPR		
001.02.005	Line-up VEX system to smoke clear	MCC/SCC WKPR		
001.02.006	Line-up VEX system to surface	MCC/SCC WKPR		
001.02.007	Secure the VEX system IAW VCSOP's	MCC/SCC WKPR		
EO 001.03Operate the AV&B System				
001.03.001	Line up the AV&B supply to the ship's whistle	MCC/SCC WKPR		

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PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
001.03.002	Isolate the Fuel Firesafe valves using manual vent valves	MCC/SCC WKPR		
EO 001.04 Operate the Voice Pipe System				
001.04.001	Prepare Voice Pipe for operation	MCC/SCC WKPR		
EO 001.05 Operate the Conning Tower Flood/Drain System				
001.05.001	Explain the procedure for draining down the conning tower	MCC/SCC WKPR		
EO 001.06 Inspect Main Battery				
001.06.001	Conduct a battery dip	MCC/SCC WKPR (Elect)		
001.06.002	Log battery dip in battery log and on state boards	MCC/SCC WKPR (Elect)		
001.06.003	Report state of battery to the OOW and CO	MCC/SCC WKPR (Elect)		
001.06.004	Describe actions upon finding moisture in battery sumps	MCC/SCC WKPR (Elect)		
001.06.005	Operate battery sump suction IAW VCSOP's	MCC/SCC WKPR (Elect)		
001.06.006	Operate battery ventilation pneumatic (PV) valves in all modes	MCC/SCC WKPR (Elect)		
EO 001.07Operate Masts and Periscopes				
001.07.001	Operate masts and periscopes using local hydel	MCC/SCC WKPR		
001.07.002	Operate masts using TTVC	MCC/SCC WKPR		
001.07.003	Conduct greasing routine on the periscopes	MCC/SCC WKPR		
001.07.004	Inspect induction and exhaust masts drain cans and explain their purpose	MCC/SCC WKPR		
EO 001.08 Operate Buoyant Wire Aerial (BWA)				
001.08.001	Conduct SOP line-up on the BWA	MCC/SCC WKPR		
001.08.002	Assist in the deployment of the BWA	Wrecker		
001.08.003	Assist in the recovery of the BWA	Wrecker		

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Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
001.08.004	Refresh the “Nip” on the BWA	Wrecker		
EO 001.09 Conduct a Set of Oncoming Rounds				
001.09.001	Conduct rounds ensuring all spaces are secured for sea	MCC/SCC WKPR		
001.09.002	Conduct rounds ensuring all firefighting gear is correct	MCC/SCC WKPR		
001.09.003	Conduct rounds ensuring all spaces clean	MCC/SCC WKPR		
001.09.004	Conduct rounds ensuring all bilge levels are correct	MCC/SCC WKPR		
001.09.005	Conduct rounds ensuring all replenishment tank levels are correct	MCC/SCC WKPR		
001.09.006	Conduct rounds ensuring all drain can levels are correct	MCC/SCC WKPR		
001.09.007	Conduct rounds ensuring all spaces clean	MCC/SCC WKPR		
001.09.008	Identify the submarines current machinery state	MCC/SCC WKPR		
001.09.009	Ensure all machinery logs are complete	MCC/SCC WKPR		
EO 001.10 Operate the Surveillance System				
001.10.001	Monitor Surveillance from an LSU	MCC/SCC WKPR		
001.10.002	Inhibit/enable a channel on surveillance	MCC/SCC WKPR		
001.10.003	Identify inhibited channels	MCC/SCC WKPR		
001.10.004	Create an SCC/MCC Operator Assist page	MCC/SCC WKPR		
EO 001.11 Operate Trim, Bilge and Ballast System Domestic Services				
001.11.001	Operate the HP Bilge and Ballast priming system cross-connection	MCC/SCC WKPR		
001.11.002	Operate the HP Bilge/Ballast pumps locally using starter and Field Regulator Local Indicator Panel (FRLIP)	MCC/SCC WKPR		
001.11.003	Prime the LP Bilge Pump	MCC/SCC WKPR		

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Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
001.11.004	Prime the Trim Pumps	MCC/SCC WKPR		
001.11.005	Prime the HP Bilge/Ballast Priming Pump Sealing Tanks	MCC/SCC WKPR		
EO 001.12 Operate Reverse Osmosis Desalinator (ROD)				
001.12.001	Conduct SOP line-up on the ROD	ROD Maintainer		
001.12.002	Start/stop the ROD	ROD Maintainer		
001.12.003	Complete a set of rounds and log in log book	ROD Maintainer		
001.12.004	Secure the ROD for domestic routines	ROD Maintainer		
001.12.005	Test a water sample for chlorination an pH level	ROD Maintainer		
PO 002 Perform Submarine Rounds Aft				
EO 002.01 Inspect Diesel Systems				
002.01.001	Complete pre-start checks on the engine	Horse		
002.01.002	Complete engine rounds while running	Horse		
002.01.003	Record rounds in the engine log book	Horse		
002.01.004	Conduct Kittiwake lube oil viscosity test	Horse		
002.01.005	Conduct Kittiwake lube oil water-in-oil content	Horse		
002.01.006	Conduct Kittiwake lube oil TBN number	Horse		
002.01.007	Conduct Kittiwake insoluble content	Horse		
002.01.008	State the frequency that each Kittiwake test is conducted	Horse		
002.01.009	Assist MCC1 in changing over fuel from Coalescer to Centrifuge	Horse		
EO 002.02Transfer Lubrication Oil to Equipment				

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Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
002.02.001	Conduct SOP line-up on the Engine Lube Oil Transfer System	MCC/SCC WKPR		
002.02.002	Transfer oil from the Clean Lube Oil Tank to the Diesel engines sumps	MCC/SCC WKPR		
002.02.003	Transfer oil from the Clean Lube Oil Tank to the Generator pedestal bearings	MCC/SCC WKPR		
002.02.004	Transfer oil from the Clean Lube Oil Tank to the Main Motor bearings	MCC/SCC WKPR		
002.02.005	Transfer oil from the Clean Lube Oil Tank to the Thrust block	MCC/SCC WKPR		
002.02.006	Transfer oil from the Clean Lube Oil Tank to the Centrifuge	MCC/SCC WKPR		
002.02.007	Transfer oil from the Clean Lube Oil Tank to the Engine governor	MCC/SCC WKPR		
002.02.008	Transfer oil from the Clean Lube Oil Tank to the LP Blower	MCC/SCC WKPR		
002.02.009	Add oil to the HPAC	MCC/SCC WKPR		
002.02.010	Operate the Auto Greaser	MCC/SCC WKPR		
002.02.011	Refill the Auto Greaser	MCC/SCC WKPR		
EO 002.03 Transfer Hydraulic Oil to Hydraulic Plants				
002.03.001	Conduct SOP line-up on the Hydraulic Transfer System	MCC/SCC WKPR		
002.03.002	Transfer hydraulic oil from the Hydraulic Storage tank to the Hydraulic Rep tanks using the hydraulic transfer system	MCC/SCC WKPR		
EO 002.04 Operate HP Air System				
002.04.001	Start-up/Shut-down HP air compressor	MCC/SCC WKPR		
002.04.002	Switch over air dryer IAW VCSOP's	MCC/SCC WKPR		
002.04.003	Recharge the SCBA bottles from the SCBA charging panel	MCC/SCC WKPR		

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PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
002.04.004	Charge the main bottle groups from ashore	MCC/SCC WKPR		
EO 002.05 Inspect Fresh Water and Chilled Water Cooling				
002.05.001	Conduct SOP line-up on Chilled Water system	MCC/SCC WKPR		
002.05.002	Conduct SOP line-up on Freshwater Cooling system	MCC/SCC WKPR		
002.05.003	Start/Stop Chilled Water Plants	MCC/SCC WKPR		
002.05.004	Conduct rounds of Chilled Water System and enter into log book	MCC/SCC WKPR		
002.05.005	Top up Chilled Water expansion tank	MCC/SCC WKPR		
002.05.006	Adjust Chilled Water flow rates	MCC/SCC WKPR		
002.05.007	Operate system and pump vents	MCC/SCC WKPR		
002.05.008	Operate ship's ventilation Air Treatment Unit (ATU) plenum cross connection	MCC/SCC WKPR		
002.05.009	Operate duct heaters in all modes	MCC/SCC WKPR		
002.05.010	Operate ATU chilled water supply solenoid valves	MCC/SCC WKPR		
002.05.011	Top-up Fresh Water Cooling Expansion Tanks	MCC/SCC WKPR		
PO 003 Perform Submarine Rounds Alongside				
EO 003.01Safety and Environment Management System				
003.01.001	Explain the purpose of the Victoria Class Safety and Environmental Management System	Wrecker		
003.01.002	Explain Due Diligence	Wrecker		
003.01.003	Demonstrate the ability to test leads of a Digital Volt Meter	MCC/SCC WKPR (Elect)		
003.01.004	Demonstrate the ability to test leads of a Digital Volt Meter	MCC/SCC WKPR (Elect)		

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PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
003.01.005	Demonstrate the ability to measure resistance with a Digital Volt Meter	MCC/SCC WKPR (Elect)		
003.01.006	Demonstrate the ability to measure AC and DC voltage using a Digital Volt Meter	MCC/SCC WKPR (Elect)		
003.01.007	Demonstrate the ability to conduct insulation resistance	MCC/SCC WKPR (Elect)		
003.01.008	Assist with an Electrical Lockout/Tagout IAW SEMS	MCC/SCC WKPR (Elect)		
003.01.009	Assist with a Mechanical Lockout/Tagout IAW SEMS	MCC/SCC WKPR (Mech)		
EO 003.02 Fueling and Transferring of POLs				
003.02.001	Participate as a member of a fueling party (valve sentry, fueling point sentry)	Fueling I/C		
003.02.002	Identify the members of a submarine fueling party.	Fueling I/C		
003.02.003	Assist with embarkation of Lube oil	BLPO		
003.02.004	Assist with embarkation of Hydraulic oil	BLPO		
003.02.005	Transfer LP Bilge tank contents ashore, IAW SEMS	BLPO		
EO 003.03 Inspect Battery Systems				
003.03.001	State all precautions prior to entering a battery compartment	MCC/SCC WKPR (Elect)		
003.03.002	Assist in main battery agitation test	MCC/SCC WKPR (Elect)		
003.03.003	Assist in main battery cell top cleaning	MCC/SCC WKPR (Elect)		
003.03.004	Assist in main battery terminal connection check	MCC/SCC WKPR (Elect)		
003.03.005	Assist in main battery electrolyte top-up	MCC/SCC WKPR (Elect)		
003.03.006	Input data into battery maintenance log	MCC/SCC WKPR (Elect)		
003.03.007	Act as Hydrogen Eliminator roundsperson during a gassing charge	MCC/SCC WKPR (Elect)		

Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
003.03.008	Test demineralized water prior to embarking	MCC/SCC WKPR (Elect)		
003.03.009	Embark/disembark demineralized water	MCC/SCC WKPR (Elect)		
003.03.010	Add water to Battery Cooling Expansion Tank	MCC/SCC WKPR (Elect)		
003.03.011	Line-up system for main battery electrolyte top-up	MCC/SCC WKPR (Elect)		
003.03.012	Conduct SOP line-up on the Battery Cooling system	MCC/SCC WKPR (Elect)		
003.03.013	Conduct SOP line-up on the Battery Ventilation system	MCC/SCC WKPR (Elect)		
EC 003.04 Inspect Lighting Systems				
003.04.001	Confirm power supplies to lighting circuit	MCC/SCC WKPR (Elect)		
003.04.002	Replace components of lighting circuit	MCC/SCC WKPR (Elect)		
003.04.003	Perform function test of lighting circuit	MCC/SCC WKPR (Elect)		
EC 003.05 Inspect Motor and Generators				
003.05.001	Confirm motor operation	MCC/SCC WKPR (Elect)		
003.05.002	Confirm motor start permissives	MCC/SCC WKPR (Elect)		
003.05.003	Confirm motor trips	MCC/SCC WKPR (Elect)		
003.05.004	Check power at motor feeder panel using DVM	MCC/SCC WKPR (Elect)		
003.05.005	Check power at motor input using DVM	MCC/SCC WKPR (Elect)		
003.05.006	Check secondary power of control transformer using DVM	MCC/SCC WKPR (Elect)		
003.05.007	Confirm status of control fuses	MCC/SCC WKPR (Elect)		
003.05.008	Confirm status of overload protections	MCC/SCC WKPR (Elect)		
003.05.009	Confirm status of Motor Thermal Protection Unit (MTPU)	MCC/SCC WKPR (Elect)		

Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
EO 003.06Inspect Bulk Liquid Storage				
003.06.001	Dip M port tank	BLPO		
003.06.002	Dip M starboard tank	BLPO		
003.06.003	Dip R port tank	BLPO		
003.06.004	Dip R starboard tank	BLPO		
003.06.005	Dip port aft trim tank	BLPO		
003.06.006	Dip starboard aft trim tank	BLPO		
003.06.007	Dip port fwd trim tank	BLPO		
003.06.008	Dip starboard fwd trim tank	BLPO		
003.06.009	Dip port WRT tank	BLPO		
003.06.010	Dip starboard WRT tank	BLPO		
003.06.011	Dip port TOT	BLPO		
003.06.012	Dip starboard TOT	BLPO		
003.06.013	Dip port domestic fresh water tank	BLPO		
003.06.014	Dip starboard domestic fresh water tank	BLPO		
003.06.015	Dip clean bilge tank	BLPO		
003.06.016	Dip dirty bilge tank	BLPO		
003.06.017	Dip clean oil tank	BLPO		
003.06.018	Dip dirty oil tank	BLPO		
003.06.019	Dip distilled water tank	BLPO		

Trainee Rank:_____Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
003.06.020	Identify tanks that have AV&B supplied blows	BLPO		
003.06.021	Locate tank blow reducers	BLPO		
003.06.022	Locate tank blow/vent valves	BLPO		
003.06.023	State pressures of blows	BLPO		
EO 003.07 Connecting Submarine to Shore Power				
003.07.001	Test shore supply cable insulation using 1000V insulation meter	MCC/SCC WKPR (Elect)		
003.07.002	Inspect/hull pen prior to cable connection	MCC/SCC WKPR (Elect)		
003.07.003	Connect shore supply cable to submarine	MCC/SCC WKPR (Elect)		
003.07.004	Reset battery switchboard links	MCC/SCC WKPR (Elect)		
EO 003.08 Inspect Domestic Services				
003.08.001	Fill FW system	BLPO		
003.08.002	Change over FW tanks	BLPO		
003.08.003	Blow off FW system	BLPO		
003.08.004	Reset FW heaters	BLPO		
003.08.005	Blow sewage tank to shore IAW VCSOP's and SEMS	MCC/SCC WKPR		
003.08.006	Conduct halocarbon leak checks	Fridge King		
003.08.007	Start/Stop fridge compressor	Fridge King		
003.08.008	Reset HP trip	Fridge King		
003.08.009	Log rounds in log book	Fridge King		

PERFORMANCE CHECK – PC 001

ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine.

Arrangements and Procedures

1. PC 001 will be a practical test based on content from PO 001- PO 003.
2. Pass / Fail.
3. Time required: Up to 6 hours at the discretion of the assessors.
4. References will not be required.
5. Conducted on completion of OJPR Annex D.

INSTRUCTIONS GIVEN TO TRAINEE

PC 001 will be a practical assessment based on content from PO 001 – PO 003.

It will be conducted in two parts, one with the Senior Mar Tech – Mech. And one with the Senior Mar Tech – Elect.

1. During this practical assessment, you will be required to conduct a complete set of rounds as an SCC2/MCC3 Roundsperson. You must pass this PC to pass PO 001 – PO 003. To pass, you must complete all performance in accordance with the standard.
2. References will not be provided.
3. Pass / Fail.
4. Do you have any questions before you start?

PERFORMANCE CHECK NOTES

This checklist is not comprehensive of what to ask the candidate. It provides the minimum standard to assess the viability of the candidate.

PERFORMANCE CHECK LIST

Item	PC 001 Performance Statement	SAT	UNSAT	Remarks
Part 1. Victoria Class Roundsperson – Mechanical Rounds				
a.	**Safety and Environmental Management System**			
	Explain the purpose of the Vic Class SEMS.	☐	☐	
	Explain the meaning of due diligence.	☐	☐	
b.	**Conduct a set of Oncoming Rounds**			
	Ensure all spaces are secure for sea.	☐	☐	
	Ensure all firefighting gear is correct.	☐	☐	
	Ensure all spaces are clean.	☐	☐	
	Ensure all bilge levels are correct.	☐	☐	
	Ensure all drain can levels are correct.	☐	☐	
	Identify the submarine's current machinery state.	☐	☐	
	Ensure all machinery logs are complete.	☐	☐	
c.	**Operate the Induction System IAW VCSOP's**			
	Line-up to flood upon diving	☐	☐	
	Drain down for Snorting	☐	☐	
	Secure under normal circumstances	☐	☐	
d.	**Operate the LP Blow System IAW VCSOP's**			
	Conduct pre-start checks of LP Blower.	☐	☐	
	Line up VEX System for pre-start checks.	☐	☐	
e.	**Operate the AV&B System IAW VCSOP's**			
	Line up air supply to Ship's Whistle	☐	☐	
	Isolate the Fuel Firesafe Valve using manual vent valve.	☐	☐	
f.	**Operate the Conning Tower System IAW VCSOP's**			
	Explain the procedure for draining down the conning tower.	☐	☐	
g.	**Operate the Masts and Periscopes**			
	Operate the masts and periscopes using local hydel.	☐	☐	
	Operate the masts and periscopes using TTVC.	☐	☐	
h.	**Operate the Trim, Bilge and Ballast Systems.**			
	Cross-connect the Bilge and Ballast Priming Pumps.	☐	☐	
	Operate the Bilge Pump locally by FRLIP.	☐	☐	

i. Operate the ROD			
Conduct a set of rounds on the ROD	☐	☐	
Log the ROD	☐	☐	
j. Inspect Diesel Systems			
Complete pre-start checks on the Diesel.	☐	☐	
k. Transfer Lubrication Oil to Equipment			
State what equipment requires oil and the type of oil required in each.	☐	☐	
Transfer 10 litres of oil to an engine sump.	☐	☐	
Refill the Auto-lube reservoir.	☐	☐	
l. Operate HP Air System			
Charge the HP Air	☐	☐	
Switch over air dryers	☐	☐	
m. Inspect Fresh Water and Chilled Water Cooling			
Top up the FW Cooling Header Tanks	☐	☐	
Start an AC Chiller.	☐	☐	
Log AC Chiller	☐	☐	
n. Fueling and Transferring of POLs			
Identify the members of the Fuelling party.	☐	☐	
State the precautions to be taken prior to transferring Bilge tanks ashore.	☐	☐	
o. Inspect the Bulk Liquid Storage			
Dip and state the levels of three different storage tanks.	☐	☐	
p. Inspect Domestic Services			
Conduct leak check of fridge compartment.	☐	☐	
Log the fridge compressor.	☐	☐	

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Positive Feedback		**Improvement Feedback**	
Part 1 Assessment – Circle one.			
Met Standard (Sat on 100% of items)		**Did Not Meet Standard**	
Assessor I have discussed the results of Part 1 with the candidate.		**Candidate** I have been debriefed on the result of Part 1.	
Part 1 Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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Item	PC 001 Performance Statement	SAT	UNSAT	Remarks
Part 2. Victoria Class Roundsperson - Electrical Rounds				
a.	**Safety and Environmental Management System**			
	Demonstrate the use of a Digital Volt Meter (DVM).	☐	☐	
b.	**Inspect Battery Systems**			
	Demonstrate all precautions prior to entering the battery.	☐	☐	
c.	**Inspect Main Battery**			
	Perform a battery dip and log in battery log.	☐	☐	
	Inspect the battery sumps and describe actions upon finding moisture.	☐	☐	
d.	**Operate the Surveillance System**			
	Monitor surveillance from an LSU.	☐	☐	
	Create an Op Assist Page.	☐	☐	
e.	**Inspect Lighting Systems**			
	Troubleshoot ship's service lighting.	☐	☐	
	Trouble shoot Navigational Lighting	☐	☐	
f.	**Inspect Motors and Generators**			
	Confirm operation of a DC Motor	☐	☐	
	Confirm power of AC motor using DVM.	☐	☐	
	Confirm status of protective devices.	☐	☐	
g.	**Connecting Submarine to Shore Power**			
	Describe how to connect AC/DC shore supply cables.	☐	☐	

Positive Feedback	Improvement Feedback

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Part 2 Assessment – Circle one.			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of Part 2 with the candidate.		**Candidate** I have been debriefed on the result of Part 2.	
Part 2 Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 001 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Administrator I have discussed the results of PC 001 with the candidate.		**Candidate** I have been debriefed on the result of PC 001.	
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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ANNEX E – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
A/C	Air Conditioner
ACOS	Automatic Change Over Switch
AC	Alternate Current
AFFF	Aqueous Film Forming Foam
B/W	Black Water
CBRN	Chemical Biological Radiological Nuclear
D/C	Damage Control
DC	Direct Current
F/W	Fresh Water
FA	First Aid
FOS	Fuel Oil System
FOT	Fuel Oil Transfer
G/W	Grey Water
GT	Gas Turbine
HMI	Human Machine Interface
HP	High Pressure
HVAC	Heat and Ventilation System
LP	Low Pressure
MCOS	Manual Change Over Switch
MLO	Main Lube Oil

PDE	Pielstick Diesel Engine
PG&D	Power Generation & Distribution
PLC	Programmable Logic Controller
S/W	Sea Water
SFC	Static Frequency Converter
SSS	Synchronous Self Shifting
TAU	Twin Agent Unit
UPS	Uninterruptable Power Supply
W/T	Watertight Door

ANNEX F – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

ANNEX G – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).

As an OJPR, no additional costing is required.

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

ANNEX H – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex E

ANNEX I – ASSESSMENT PLAN

Covered in Annex D – OJPR